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Bayesian Surprise Predicts Human Event Segmentation in Story Listening

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Abstract

Event segmentation theory posits that people segment continuous experience into discrete events and that event boundaries occur when there are large transient increases in prediction error. Here, we set out to test this theory in the context of story listening, by using a deep learning language model (GPT-2) to compute the predicted probability distribution of the next word, at each point in the story. For three stories, we used the probability distributions generated by GPT-2 to compute the time series of prediction error. We also asked participants to listen to these stories while marking event boundaries. We used regression models to relate the GPT-2 measures to the human segmentation data. We found that event boundaries are associated with transient increases in Bayesian surprise but not with a simpler measure of prediction error (surprisal) that tracks, for each word in the story, how strongly that word was predicted at the previous time point. These results support the hypothesis that prediction error serves as a control mechanism governing event segmentation and point to important differences between operational definitions of prediction error.

Keywords: Event segmentation; GPT-2; Surprise; Entropy; Bayesian surprise; Narratives

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1. Introduction

A large body of work has established that people make sense of continuous experience by parsing it into discrete events (for reviews, see Clewett, DuBrow, & Davachi, 2019; Zacks, 2020; Zacks et al., 2001); this holds for audiovisual (Baldassano et al., 2017; Ben-Yakov & Henson, 2018; Geerligs, Gerven, Campbell, & Güçlü, 2021; Zacks, Speer, & Reynolds, 2009), auditory (Michelmann et al., 2021; Zacks et al., 2009), and text stimuli (Ezzyat & Davachi, 2011; Pettijohn & Radvansky, 2016; Zacks et al., 2009). For example, studies have found that people show substantial agreement regarding the timing of event boundaries—moments when one event ends and the next event begins (Magliano, Radvansky, Forsythe, & Copeland, 2014; Sargent et al., 2013; Saylor, Baldwin, Baird, & LaBounty, 2007; Sherrill, Kurby, Lilly, & Magliano, 2019; Swallow, Kemp, & Candan Simsek, 2018; Swallow & Wang, 2020; Zacks, Speer, Vettel, & Jacoby, 2006). Furthermore, this process of event segmentation has been shown to have strong psychological consequences. For example, both event recall and recognition memory correlate with event segmentation across the lifespan (Bailey et al., 2013; Sargent et al., 2013). Event segmentation also predicts action performance on typical daily living tasks such as packing a lunch box (Bailey, Kurby, Giovannetti, & Zacks, 2013). People are better at remembering the temporal order of items within events than across events (e.g., Clewett, Gasser, & Davachi, 2020; DuBrow & Davachi, 2013, 2014, 2016; Ezzyat & Davachi, 2011; Heusser, Ezzyat, Shiff, & Davachi, 2018; Horner, Bisby, Wang, Bogus, & Burgess, 2016; Rouhani, Norman, Niv, & Bornstein, 2020; Sols, DuBrow, Davachi, & Fuentemilla, 2017) and remember items presented at event boundaries better, for the most part, than within-event items (e.g., Newton & Engquist, 1976; Sonne, Kingo, & Krøjgaard, 2017; Swallow et al., 2011; Swallow, Zacks, & Abrams, 2009). Event boundaries also affect the contents of working memory—for example, Swallow et al. (2011) found that brain systems involved in long-term memory were engaged when remembering items presented 5 s ago that were part of the previous event, more so than when remembering items presented 5 s ago that were part of the current event; they explain this by arguing that items that are part of the current event are still in an active state in memory (and thus do not need to be retrieved from long-term memory).

Event segmentation theory (EST; Zacks, Speer, Swallow, Braver, & Reynolds, 2007) seeks to account for these diverse findings by positing that people maintain a multimodal representation of the current event's attributes—a *working event model*—that is used to predict what will happen next in the event, thus facilitating our processing of the next moment. According to EST, the working event model is held in an active state in memory as the event progresses so long as its predictions are met. However, once mismatches start to occur between the working event model's predictions and the sensory environment (i.e., *prediction errors*), it is no longer adaptive to keep this working event model active. The working event model is then updated with information about the new event that is unfolding. These moments, when there are transient increases in prediction error and the working event model is updated, constitute event boundaries, signaling the end of one event and the beginning of a new event. Thus, a key assertion of EST is that an increase in prediction errors leads to an event boundary (Zacks et al., 2007).

The predictions of EST have been validated in a variety of paradigms using both behavioral and neural measures of prediction error (e.g., Antony et al., 2021; Zacks, Kurby, Eisenberg, & Haroutunian, 2011; Zacks et al., 2009). For example, Zacks et al. (2011) showed participants movies of people performing everyday tasks, such as washing a car or doing laundry; participants were interrupted at various points in the movie and asked to predict a scene that would occur 5 s later. Zacks et al. (2011) found that participants had greater prediction accuracy when the scene that they were asked to predict was within the same event that they were currently viewing, compared to when they made predictions about a time point that belonged to the following event (about two and a half seconds after the point of interruption). This study also showed increased brain activity for predictions across event boundaries compared to predictions within event boundaries in the right substantia nigra (a region that is a part of the midbrain dopaminergic system that responds preferentially to prediction errors; Aron et al., 2004).

In a study of people viewing basketball games, where beliefs about which team is winning undergo frequent changes (Antony et al., 2021), prediction error was operationalized as the change from one moment to the next in the probability of a team winning the game. The probability of a team winning at each moment was mathematically modeled using factors such as the points scored and the remaining time in the game. The change in probability of a team winning, termed surprise, was computed for each moment of the game. Event boundaries tended to occur at moments when the surprise was large. Furthermore, this relationship was strongest for moments when participants believed that one team was going to win, but the game situation changed and the probability of the other team winning increased—termed *belief-inconsistent surprise*. Belief-inconsistent surprise was also correlated with neural event boundaries, measured using a data-driven Hidden Markov Model (Baldassano et al., 2017) that was applied to functional magnetic resonance imaging (fMRI) data collected during viewing of the basketball games. In another study, Zacks et al. (2009) used a reading paradigm and estimated the predictability of word clauses using an independent group of participants; they found that the predictability of a clause was inversely related to event segmentation—greater clause predictability was associated with a decreased probability that an event boundary occurred in that clause. Thus, the key claim of EST, that prediction error leads to event boundaries, finds consistent support across a wide range of paradigms and measurement techniques. Note that judgments of prediction error can also be computed as a backward inference, where sudden discontinuities in the experience indicate that the current event ended at the prior time point (Papenmeier, Brockhoff, & Huff, 2019).

While the aforementioned results are generally consistent with EST, these results are subject to several limitations. Most of these studies (all except Antony et al., 2021) lack an objective measure of prediction error that can be computed on a moment-by-moment basis. For example, ratings of predictability that are retrospectively collected at the end of an experiment (Zacks et al., 2009) are an imperfect measure of prediction—something can seem predictable even if it was not predicted when performing the task. Forced-choice prediction judgments (e.g., Zacks et al., 2011) are vulnerable to similar effects as participants can choose a picture as being more likely to occur even if they did not predict it. Frequently interrupting the narrative to get predictability judgments (as in Zacks et al. 2011) is also not ideal, as it interferes

with the online processing of narratives. Another issue with both of these methods is that they only measure prediction at a limited number of time points. Ideally, we would like to collect a continuous time series of prediction, and then compare that on a fine-grained basis with observed data, to compute a time series of prediction error. This prediction error could then be related to segmentation ratings.

There are several ways around these drawbacks, each with their own limitations. One could use continuous implicit measures like eye movements to track predictions, as in Eisenberg, Zacks, and Flores (2018). However, the trajectory of eye gaze may not always pertain to prediction, as eye gaze is also influenced by salience in the image (Peters, Iyer, Itti, & Koch, 2005). Also, eye tracking can only be used to reveal the participant's predictions when the stimuli are movies or still images; it cannot be used for story listening or text stimuli. For stories, we could collect norming data about the predictability of the next word from a separate set of participants. Numerous studies have used this technique to get word probabilities for sentence-ending words (*cloze* probabilities; Kutas & Hillyard, 1980); for example, given the frame "I drink my coffee with cream and," the completion "sugar" is provided very frequently (leading to a high *cloze* probability), and the completion "dog" is provided very infrequently (leading to a low *cloze* probability). This approach was recently extended by Goldstein et al. (2022) to get next-word predictions after every word (not just sentence-ending words) for an entire podcast (see also Michelmann et al., 2021). The limitation of this approach is that it is extremely labor intensive, even with online data collection. Using mathematical models to compute prediction error could overcome these limitations of using participants to norm data. For example, in the Antony et al. (2021) study, the prediction error time series was related to neural and behavioral measures of event segmentation. The results were promising, but this method can only be applied in a limited domain (basketball games) to predict a particular variable (game outcome). Ideally, if we could model the next-moment probabilities in a wider range of situations, not just for games, this would provide a more powerful platform for testing EST.

1.1. GPT-2

In the present study, we address the above limitations by using GPT-2 (Radford et al., 2019), a recently developed large-scale neural network model of language to generate a time series of predictions for text stimuli (for a related approach that has been applied to video stimuli, see Fountas et al., 2022). The so-called *transformer* models like GPT-2 learn to perform sophisticated transformations on a large trailing window of words to predict held-out text (this trailing window, called the "context window," can be hundreds of words in size). Here, we use GPT-2 (Radford et al., 2019) to generate next-word predictions on a word-by-word basis from stories; from these predictions, we compute various measures of prediction error as described in the next section. In practice, the transformer architecture has revolutionized natural language processing, overcoming the limitations of previous architectures that used a much narrower context window to make predictions (see, e.g., Armeni, Willems, & Frank, 2017). Transformer models like GPT-2 outperform previous models in key benchmarks, including next-word prediction (Devlin, Chang, Lee, & Toutanova, 2019; see also Dai

et al., 2019). Also, although GPT-2 does not fully mimic human abilities in language comprehension (Bender & Koller, 2020), recent work has shown that model-derived word probabilities correlate with human judgments of surprise (Goldstein et al., 2022) and with neural activity related to language comprehension (Goldstein et al., 2022; Heilbron, Armeni, Schoffelen, Hagoort, & Lange, 2020; Schrimpf et al., 2021; Toneva, Mitchell, & Wehbe, 2022).

1.2. Testing theories of event segmentation

The rich time series of predictions generated by GPT-2 provides an unprecedented opportunity for quantitatively comparing different theories of what drives event segmentation.¹ In this study, we used GPT-2 to compute four distinct measures of prediction error and its correlates, herein called *disfluency* measures and assessed how strongly each of these disfluency measures was associated with human-annotated event boundaries.

The first disfluency measure we explored was *surprisal* (Armeni et al., 2017; Eisape, Zaslavsky, & Levy, 2020; Hale, 2001, 2016; Kuperberg, 2021; Kuperberg & Jaeger, 2016). We computed a time series of surprisal by measuring, for each word in the story, how strongly that word was predicted by GPT-2 at the preceding time point (i.e., after the immediately-preceding word was presented); we then computed the negative log of each of these predicted probabilities. The effect of this negative log transformation is that words in the story that are less strongly predicted are associated with higher values of surprisal. This way of operationalizing prediction error has some features in common with the approach of computing cloze probability (the probability of sentence-ending words) from norming data (see, e.g., Eisape et al., 2020; Federmeier, 2007; Kuperberg & Jaeger, 2016; Kutas & Hillyard, 1980; Szwedczyk & Federmeier, 2022). Some key differences between our approach and the cloze approach are (1) our surprisal measure is computed automatically by GPT-2 (no extra norming data are required); (2) our surprisal measure is generated for all words in the story, whereas cloze has typically been computed for sentence-ending words; and (3) our surprisal measure factors in a larger amount of context than the current sentence (due to GPT-2's capacity to hold a large trailing window of words in its memory and use these words as the context for the next word prediction).²

Importantly, surprisal is not the only way to operationalize prediction error. A second approach is to compute the change in your overall belief state that occurs as a result of a word being presented. Intuitively, a word that changes your overall belief state substantially is more surprising (i.e., more belief inconsistent) than one that only changes your belief state by a small amount. The Antony et al. (2021) study described above operationalized surprise as a change in belief state, but only in a very limited way (who will win the basketball game). For narrative prediction, we can measure the change in belief state in a more general way by looking at the change (from time point to time point) in the overall probability distribution over words. That is, we can treat the probability distribution over words at time $t - 1$ as a prior and look at how the prior distribution is updated by the observation of a new word at time t . This difference between a prior distribution and the current distribution is referred to as *Bayesian surprise* and is measured by computing the Kullback–Leibler divergence (KLD) between the two probability distributions (Itti & Baldi, 2009; for applications to language comprehension,

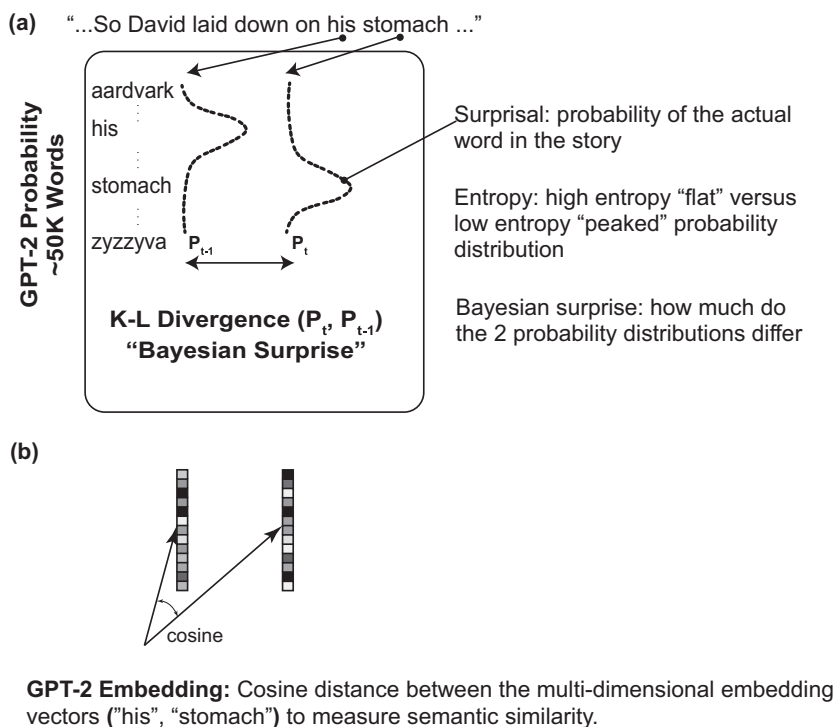


Fig. 1. (a) Schematic showing the GPT-2 probability distributions at two word points and the derived measures of surprisal, entropy, and Bayesian surprise (for the mathematical definitions of these measures, see Section 2). (b) The cosine distance between successive word embedding vectors extracted from the last hidden layer of GPT-2. The smaller the embedding cosine distance, the greater the semantic similarity between words.

see Kuperberg, 2021; Kuperberg & Jaeger, 2016; Lindborg, Musiolek, Ostwald, & Rabovsky, 2022; Rabovsky, Hansen, & McClelland, 2018; Rabovsky & McRae, 2014; for applications to video prediction, see Fountas et al., 2022; for applications to memory, see Quent, Henson, & Greve, 2021; Henson & Gagnepain, 2010; for fMRI and pupillometry data dissociating Bayesian surprise from surprisal, see O'Reilly et al., 2013).

As illustrated in Fig. 1, Bayesian surprise factors in the shape of the entire probability distribution, whereas surprisal only factors in the probability of the next word. One key limitation of surprisal, relative to Bayesian surprise, is that surprisal is not sensitive to the confidence of the next-state prediction (Baldwin & Kosie, 2021; Richmond & Zacks, 2017). Consider the following two scenarios: one where you are certain about what word will come next, but the prediction was wrong, leading you to become uncertain; and another where you are uncertain and that uncertainty persists. The former scenario aligns better with the intuitive concept of a prediction error, insofar as it reflects having a wrong prediction, rather than simply having no prediction. This difference is not captured by surprisal (since the predicted probability of the actual next word is low in both cases), but it is successfully captured by Bayesian surprise (since the overall shape of the predictive distribution changes substantially in the former

scenario—going from peaked to flat—but not the latter scenario). These ideas align with prior work suggesting that “unexpected uncertainty” (i.e., a shift from certainty to uncertainty) is associated with changes in the current context (Dayan & Yu, 2003; Kuperberg & Jaeger, 2016; Nassar, Wilson, Heasley, & Gold, 2010; Yu & Dayan, 2005) and drives learning agents to seek out a new explanation for these observations (i.e., a new event model). The fact that Bayesian surprise is sensitive to the “expectedness” of uncertainty (but surprisal is not), therefore, suggests that it might do a better job of predicting event boundaries than surprisal.

In addition to comparing surprisal and Bayesian surprise (both of which can be viewed as measuring prediction error), we also compared these two measures to two additional disfluency measures that do not track prediction error per se but are related to prediction error. The first of these additional disfluency measures, *entropy*, is an information-theoretic measure of uncertainty that has been used extensively in next-word prediction models (Armeni et al., 2017; Hale, 2001, 2016; Kuperberg & Jaeger, 2016). High entropy indicates high uncertainty, and low entropy indicates low uncertainty (see Section 2 for mathematical definition). We looked at entropy here because several authors have raised the possibility that the uncertainty of the current moment could drive event segmentation (Baldwin & Kosie, 2021; Kuperberg, 2021; Richmond & Zacks, 2017). To test this, we computed a time series of uncertainty by measuring the entropy of the GPT-2-generated probability distribution of words at each time point in the story.

With our fourth and final disfluency measure, we wanted to test the hypothesis that changes in event features—such as temporal order, spatial layout, agents, intention of the agents, and causal effects—trigger event boundaries (Radvansky & Zacks, 2011; Zacks et al., 2009; Zwaan, Langston, & Graesser, 1995). In line with this hypothesis, prior work has shown that event boundaries were most likely to occur when entities changed in more than one of these dimensions (Huff, Meitz, & Papenmeier, 2014). This was true across modalities of auditory and visual presentation (Zacks et al., 2009). This kind of “feature change” can be tracked in neural network models like GPT-2 by looking at the changes in internal representations (called *embeddings*) that are evoked by each word. Word embeddings are vector-based representations of words within the internal architecture of neural networks. These vectors, which correspond to the activation of units in the network, capture emergent linguistic properties such as semantics and syntax for each word. The development of these properties occurs as the neural network undergoes training on extensive text corpora (see McClelland, Hill, Rudolph, Baldridge, & Schütze, 2020, for an overview). Embeddings from neural network models such as GloVe (Pennington, Socher, & Manning, 2014) and Word2Vec (Mikolov, Sutskever, Chen, Corrado, & Dean, 2013) have been shown to correlate with human semantic similarity judgments (Pereira, Gershman, Ritter, & Botvinick, 2016; Pereira et al., 2018). Notably, embeddings from transformer models like GPT-2 reflect not only the properties of the current word but also how the meaning of that word has been “inflected” by context—as such, changes in the overall meaning of the text from one word to the next will be reflected in changes in the (contextualized) embeddings evoked by those words (Goldstein et al., 2022; Heilbron et al., 2020). Concretely, we can compute a time series of feature change by computing, at each time point, the cosine distance between the embedding at the current time point and the embedding at the previous time point. Note that this *embedding cosine distance* measure is closely related to Bayesian surprise; the main difference is that it is tracking the

changes in the internal representations of the network, rather than changes in the predictions that the model is generating based on these internal representations.

One last dimension to consider relates to whether to normalize these measures before using them to predict event boundaries. EST posits that it is specifically transient prediction error that drives event segmentation (Zacks et al., 2007). That is, encountering a prediction error after a period of prediction errors being low (indicating good understanding) is hypothesized to be more indicative of an event boundary than encountering a prediction error after a long series of prediction errors—this latter scenario is indicative of prolonged confusion more so than the end of a well-understood event. The importance of factoring transience into prediction of event boundaries was crystallized by simulation work conducted by Reynolds, Zacks, and Braver (2007). In this simulation work, Reynolds et al. (2007) generated a sequence of artificial events and then trained a recurrent neural network to predict these events. They found that they were better able to predict the “ground truth” event boundaries using the prediction errors generated by the neural network when they computed the change in prediction error (normalizing by a running average), compared to when they looked at instantaneous prediction error. To test the importance of transience in our study, we computed transient versions of all measures (where we normalized by a running average) and also non-transient versions of these measures.

In summary: we set out to use GPT-2 to test the relative efficacy of four different metrics of disfluency for predicting event boundaries. In keeping with EST’s focus on prediction error as a determinant of event boundaries, we looked at two measures of prediction error (surprisal and Bayesian surprise); we also looked at two other metrics that are related to, but distinct from, prediction error: entropy (a measure of uncertainty) and embedding cosine distance (a measure of feature change). Lastly, we computed both “transient” versions of all of these metrics (that were normalized by a running average) and non-transient versions.

We used three auditory stories to test EST: “Monkey in the Middle” (Chavis, 2017), “Pie-man” (O’Grady, 2008), and “Tunnel Under the World” (Pohl, 1956). For each of these stories, we obtained human event segmentation data (the segmentation for “Monkey in the Middle” was newly obtained for this study; for “Pie-man” and “Tunnel Under the World,” we used segmentation data that had been obtained in other studies). To test EST, we performed a series of regression analyses where we used each of the GPT-2-derived measures of disfluency to predict the human judgments of event segmentation. Based on EST, we hypothesized that measures of transient prediction error would do the best job of predicting event boundaries; we further hypothesized that Bayesian surprise would do better than surprisal because of its ability (described above) to factor in the certainty of predictions.

2. Methods

2.1. Ethical approval

Ethical approval for the studies was granted by the Princeton University Institutional Review Board.

2.2. Narrative stimuli

We used three auditory narrative stories in our study: “Monkey in the Middle” (Chavis, 2017), 30 min long, an audio podcast on “This American Life”; “Pieman” (O’Grady, 2008), 7 and a half minutes long; and “Tunnel Under the World” (Pohl, 1956), 25 min long, broadcast on the radio show “X Minus One.” For the stories “Pieman” (Michelmann et al., 2021) and “Tunnel Under the World” (Lositsky et al., 2016), we analyzed previously published segmentation data.

2.3. Behavioral segmentation

We had participants segment the story “Monkey in the Middle” on Amazon Mechanical Turk. We recruited participants on the Cloud Research platform. Following Newtonson (1973), participants were instructed to press the space bar on their keyboard during story listening whenever they thought that one meaningful event ended and another had begun. As the story was 30 min long, and we wanted participants to be fully engaged in the task, we divided the story into roughly four equal sections. For section 1, participants segmented the story as they listened to the story. For section 2, participants first listened to section 1 without segmenting and were asked to segment section 2. Similarly for sections 3 and 4, participants first only listened to the sections before the section they were asked to segment. Each section was segmented by a different group of participants. The number of participants for each part is as follows: part 1 had 30 participants, part 2 had 35 participants, part 3 had 31 participants, and part 4 had 42 participants. The stimuli were presented using Millisecond Software (Inquisit, 2021). The participants were shown a blank white screen while they listened to the story, and a black circle was briefly shown whenever they pressed the space bar.

For “Monkey in the Middle,” we computed the proportion of participants that pressed a button, indicating an event boundary, at each time point. We marked the 2-s time interval around each button-press (1 s on each side of the button-press) as containing an event boundary. This was then averaged across all participants. Gaussian smoothing was applied to the average signal across participants using the Python package `scipy.ndimage.gaussian_filter1d` with a standard deviation of 0.2 s, corresponding to approximately a 1-s interval of points in the smoothing kernel. This averaged and smoothed signal was used for the regression analysis (described below).

For “Pieman,” we used the button-press proportions from prior work (Michelmann et al., 2021). Two hundred and five participants segmented the story twice, once while listening to the story for the first time and once while listening to it for a second time. Here, we used the segmentation data from when participants were listening to the story for the first time. For “Tunnel Under the World,” the story was segmented by 10 participants as a part of an fMRI study (Lositsky et al., 2016). Note that the segmentation for “Pieman” and “Tunnel Under the World” used the same averaging and Gaussian smoothing procedures that were used for “Monkey in the Middle” (described above).

2.4. Language model: GPT-2 embeddings

We used the pre-trained deep learning language model GPT-2 (XL version) to extract contextual embeddings for each word in the story. GPT-2 has approximately 1.5 billion

parameters and a context window of 1,024 tokens (which represent full words if the word is in the GPT-2 vocabulary, or parts of words if the word does not exist in the GPT-2 vocabulary), and uses masked self-attention (Vaswani et al., 2017). Importantly, GPT-2 (like humans) only uses tokens from previous time points to predict the current word. GPT-2 has been pre-trained on 8 million webpages, and it uses a stack of 48 decoding layers (called hidden layers) from which we can extract embeddings. The word embeddings have a dimension of $1 \times 1,600$ for each layer. Using the Hugging Face transformer library (Wolf et al., 2019), we converted each word in the story into tokens using a tokenizer in Hugging Face. The tokens were then parsed through the pre-trained GPT-2 model, with a moving context window of 1,024 tokens. For each token, we extracted the embedding of the token from the top hidden layer (as was done by Goldstein et al., 2022). GPT-2 predicts the next word using the top hidden layer; we reasoned that the top layer would therefore contain the most information about the context. Finally, we used principal component analysis (PCA) to reduce the embedding dimension from 1,600 to 50, as Goldstein et al. (2022) found that reducing the dimension led to better word prediction performance. This reduced-dimension embedding was extracted for each of the three stories in our study.

2.5. GPT-2 derived measures

2.5.1. Surprise, entropy, and Bayesian surprise

To predict the next word in a text, GPT-2 uses all the tokens in its context window (1,024 tokens) and assigns a logit value to each of the $\sim 50K$ words in its vocabulary. We computed the softmax from these logit values to get a probability distribution for all words in the GPT-2 vocabulary. This was done for each prediction starting with the first word. From these softmax values, we derived three measures: surprisal, entropy, and Bayesian surprise. Following prior work (Hale, 2016), we defined the surprisal S at word-point t as the negative log of the predicted probability, generated at word-point $t - 1$, of the word that was presented at word-point t :

$$S_t = -\log(p_w),$$

p_w = probability at word-point $t - 1$ of the word w presented at time t .

From this probability distribution, we also derived the entropy, E_t , at each word-point in the story from the predicted probability distribution generated at $t - 1$,

$$E_t = -\sum_{v=1}^V p_v * \log(p_v)$$

p_v = Probability value for word v in the vocabulary

$V \in$ all words in GPT-2 vocabulary.

Bayesian surprise is defined as the KLD between a prior distribution and a posterior distribution (Itti & Baldi, 2009). In our case, we computed the Bayesian surprise between the probability distributions for successive words in the story, treating the distribution at word-

point $t - 1$ as the prior distribution and the distribution at word-point t as the posterior distribution.

$$\text{Bayesian surprise}(P_t, P_{t-1}) = - \sum_{v=1}^V p_v * \log \left(\frac{q_v}{p_v} \right),$$

$P_{t-1}, P_t \equiv$ probability distribution at word points $t - 1, t$

$p_v =$ probability of each word in the vocabulary drawn from P_t

$q_v =$ probability of each word in the vocabulary drawn from P_{t-1}

2.5.2. Embedding cosine distance

To determine the embedding cosine distance at word-point t , we computed the cosine distance between the embedding present at word-point $t - 1$ and the embedding present at word-point t . Cosine distance measures the angle between embedding vectors; smaller values of cosine distance imply greater similarity between the embeddings than larger values of cosine distance. We used the cosine distance function from the `scipy.spatial.distance` package.

2.6. Transient increase in disfluency (Reynolds et al., 2007)

Following Reynolds et al. (2007), who normalized the prediction error with the running average of prediction error using a “drift” parameter, we used the same normalization with our measures of surprisal, entropy, Bayesian surprise, and embedding cosine distance.

We initialized the running-average measure at the first word using the average of the measure for the entire story. This average value is subsequently updated for the next word based on the difference between the measure and the average. For example, in the case of surprisal,

$$\text{Average at word 1, } \mu_1 = \frac{1}{W} \sum_{w=1}^W S_w.$$

$W =$ number of words in the story

$$\mu_2 = \mu_1 + (S_2 - \mu_1) \times \text{drift}$$

$$\mu_3 = \mu_2 + (S_3 - \mu_2) \times \text{drift}$$

$\vdots$

$$\mu_W = \mu_{W-1} + (S_W - \mu_{W-1}) \times \text{drift},$$

where $\mu_1, \mu_2, \dots, \mu_W \equiv$ Average at word 1, 2, $\dots$, W

and

$\text{drift} =$ smoothing parameter for the running average.

The transient surprisal for word at time t was defined as

$$\text{Transient_Surprisal}_t = \frac{S_t}{\mu_{t-1}}.$$

For the first word in the story, we divided by μ_1 .

To check the stability of the model for a range of parameters, we computed transient measures for three drift values,

$$\text{drift} = 0.03, 0.05, 0.10.$$

2.7. Word onset time

Our regression analyses (described in the next session) required us to specify the onset time of each word in the three stories. The word onset times for the stories “Monkey in the Middle” and “Pieman” were obtained from prior work. For the story “Tunnel Under the World,” we used the Penn forced aligner package (Yuan & Liberman, 2008) to get the onset time for each word in the story.

2.8. Regression analyses

We computed a regression model for each model-derived disfluency measure: surprisal, entropy, Bayesian surprise, and the embedding cosine distance. The model-derived measure was used as the independent variable, and the button-press proportion was the dependent variable. We used a trailing window of size N words as the features in the regression and used the LASSO method in the sklearn library. The button-press proportion values were extracted at the time points corresponding to the word onsets of the story. For the regression, we created a feature matrix from the signal, for example, Surprisal, S , with a regression window of size N words, specified as follows:

$$\text{Feature-matrix } X = \begin{bmatrix} S_1 & S_2 & \dots & S_N \\ S_2 & S_3 & \dots & S_{N+1} \\ S_3 & S_4 & \dots & S_{N+2} \\ & \vdots & & \\ S_{W-N+1} & \dots & S_W \end{bmatrix}, \text{ Button-press proportion } Y = \begin{bmatrix} bp_N \\ bp_{N+1} \\ bp_{N+2} \\ \vdots \\ bp_W \end{bmatrix}$$

where $bp_N \equiv$ the button-press proportion at word-point N .

Each row in this matrix represents a trailing window of word-points for the disfluency measure. We use this matrix to predict the corresponding button-press proportion value for the last word position in row, Y . The predicted button-press proportion was then correlated with the actual button-press proportion as described below.

The model was trained and tested using a leave-one-story-out nested cross-validation scheme, with one story used for training, another story used for validation, and the third story used for testing. This way there was no overlap in the training and test data. For each loop of the cross-validation, the best model was determined (across a range of the regularization

hyperparameter α) from the cross-validation step. The best model was then used to predict the button-press proportion for the test data. This model-predicted output was correlated (using Pearson correlation) with the actual button-press proportion for the corresponding story. Finally, the average of these correlations across the three stories was computed.

2.9. Null test

To compute the significance of the computed correlation, we compared this value with the correlations computed by circularly shifting the surprisal measures and running the same nested cross-validation regression pipeline that we used for the actual signal. We circularly shifted the signal, one word at a time, for 4,000 shifts in the positive direction and 4,000 shifts in the negative direction (8,000 shifts in total). We computed the p -value for the unshifted (actual) correlation to this set of 8,000 null correlation values by computing the percentile value of the unshifted correlation and subtracting it from 100.

To test how robust the model is to the size of the regression window, we used five regression window sizes: 20, 50, 100, 150, and 200. For each of these regression windows, we ran the entire regression pipeline and the circularly shifted null test.

2.10. Model comparison

To determine which model across *drift* and regression window parameters best explains the data with the fewest number of parameters, we computed the Akaike information criteria (AIC; Akaike, 1973). AIC evaluates model complexity and performance with a preference for the best performing model with the fewest parameters. The AIC is defined as follows:

$$\text{AIC} = l \times \log(\text{Mean Squared Error}) + 2 \times (M + 1),$$

$$l = \text{number of test samples}$$

$$M = \text{number of model parameters.}$$

We computed the mean AIC for each regression model. For regression models using transient disfluency measures, the number of model parameters was equal to the size of the regression window N plus 1 (for the drift parameter); for regression models using non-transient disfluency measures, the number of model parameters was N (as there was no drift parameter). For each testing fold in our cross-validation procedure, we computed the AIC using the mean squared error between the predicted button-press proportion and the actual value. We averaged the AIC values across the three folds and reported the mean AIC.

3. Results

We predicted human segmentation data for the three stories using four measures of disfluency: surprisal, entropy, Bayesian surprise, and embedding cosine distance. For each of these four measures, we computed regression models using leave-one-story-out cross-validation with LASSO regression (see Section 2), and we correlated the predicted output from the

Table 1
Correlation and percentile scores for transient and non-transient disfluency measures for different regression windows. The transient measures are computed for three values of the drift parameter. The values in bold denote percentile scores greater than or equal to the 95th percentile

Disfluency	Regression Window	Drift = 0.03		Drift = 0.05		Drift = 0.10		Non-Transient	
		Correlation	%	Correlation	%	Correlation	%	Correlation	%
Bayesian surprise	20	0.07	97.7	0.06	96.3	0.05	95.4	0.05	86.1
	50	0.07	97.3	0.07	95.0	0.06	96.0	0.07	91.0
	100	0.10	99.6	0.10	98.9	0.09	99.2	0.09	95.4
	150	0.10	99.5	0.11	99.7	0.11	99.8	0.11	97.5
	200	0.10	96.9	0.11	98.2	0.12	99.2	0.10	95.7
Surprisal	20	0.0	47.6	0.01	53.2	0.0	50.9	−0.03	28.2
	50	−0.06	3.6	−0.06	3.8	−0.05	7.2	−0.07	10.0
	100	−0.04	12.9	−0.04	12.4	−0.04	14.3	−0.05	12.2
	150	−0.03	17.7	−0.03	18.2	−0.04	19.6	−0.03	19.6
	200	−0.03	27.9	−0.03	28.1	−0.04	20.2	−0.02	29.6
Entropy	20	−0.02	27.8	−0.01	37.0	0.00	49.0	−0.03	29.5
	50	−0.06	5.3	−0.05	9.2	−0.04	14.2	−0.10	4.5
	100	0.00	57.2	0.01	62.7	0.02	68.6	−0.03	32.0
	150	0.01	56.6	0.01	61.7	0.02	68.4	−0.02	33.6
	200	−0.02	37.8	−0.01	47.5	0.02	68.8	−0.03	29.3
Embedding cosine	20	0.02	70.5	0.02	76.7	0.03	83.2	−0.05	17.5
	50	0.05	91.3	0.04	86.9	0.03	82.6	0.01	53.4
	100	0.06	92.0	0.05	91.1	0.04	85.2	0.01	56.8
	150	0.00	48.4	0.00	45.0	−0.01	37.0	−0.01	43.0
	200	0.02	73.7	0.03	77.2	0.03	76.4	0.02	65.2

regression with the actual button-press proportion (see Fig. S1). We explored “transient” versions of the disfluency measures (where we normalized these measures by a running average with a *drift* parameter set to 0.03, 0.05, or 0.10) and non-transient versions of these measures; we also varied the size of the trailing window for the regression.

The results of these regression analyses are shown in Table 1. Overall, we found that transient Bayesian surprise did the best job of predicting human judgments of event boundaries. Notably, transient Bayesian surprise robustly predicted event segmentation across all of the hyperparameters (*drift* values and regression window sizes) that we tried. Non-transient Bayesian surprise also performed reasonably well, but results were less good for smaller regression window sizes. None of the other three disfluency measures reliably predicted human event boundaries.

To quantitatively compare the different measures, we computed the AIC for each model variant (see Section 2). We found that the transient Bayesian surprise model with *drift* = 0.10 and a regression window of 20 words showed the best trade-off between model complexity and performance (AIC = −29, 780.1; Fig. 2). More generally, we found that the transient Bayesian surprise model with *drift* = 0.10 yielded the best (lowest) AIC score across all disfluency measures—this was true for each of the individual regression-window sizes. In most cases, the AIC differences between transient Bayesian surprise and other measures were quite substantial (on the order of 10s or 100s).

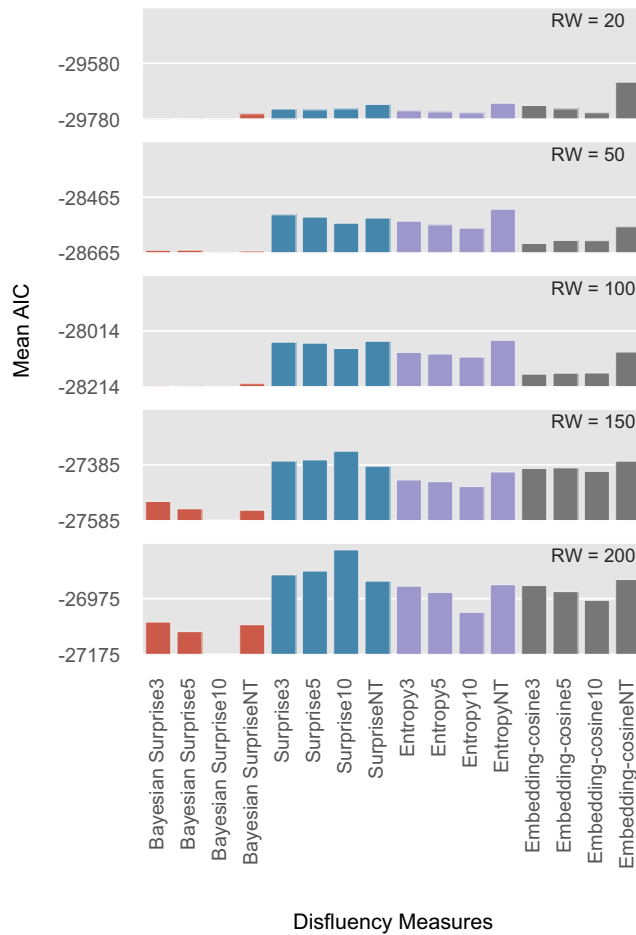


Fig. 2. The AIC scores for transient and non-transient disfluency measures, plotted separately for each regression window size. RW = regression window; the transient measures for each drift value are denoted by the suffixes (3, 5, 10) representing $drift = 0.03, 0.05, 0.10$; the non-transient measures are denoted by the suffix “NT.” To highlight differences from the lowest AIC score, the lowest y-value for each plot is the lowest AIC score for that regression window. The lowest AIC scores for each regression window are seen for transient Bayesian surprise with $drift = 0.10$; numerically, the lowest AIC scores for each of the regression windows are as follows: $AIC_{20} = -29,780.1$, $AIC_{50} = -28,664.8$, $AIC_{100} = -28,214.2$, $AIC_{150} = -27,585.0$, $AIC_{200} = -27,175.1$, where the subscript denotes the size of the regression window.

For completeness, we also checked the results for transient Bayesian surprise within each of the three stories (Table S1 in the Supporting Information). The results show that the predictive relationship went in the right direction for all three stories but was somewhat stronger for “Monkey in the Middle” and “Pieman” (where results exceeded the 95th percentile of the null for several combinations of $drift$ values and regression window sizes) as compared to “Tunnel Under the World” (where the 95th percentile of the null was only exceeded when we used $drift = 0.10$ and a regression window of size 200). In keeping with the claim from EST

that increases in disfluency at the end of an event drive event segmentation, the beta weights were high near the end of the regression window (see Fig. S2).

We also checked if our results were sensitive to the choice of smoothing parameters used to compute the button-press proportion. We computed estimates of button press using a non-parametric kernel density estimation technique (see Section 2). The results from this non-parametric estimate show a similar pattern of correlation and AIC values, with transient Bayesian Surprise showing the best correlations with button-press (Tables S2 and S3 and Fig. S3 in the Supporting Information).

To investigate why the correlations saturate with increasing regression window size, we compared the predicted button-press proportions for different regression window sizes (Fig. S4). We see that smaller window sizes show smaller amplitude fluctuations and larger window sizes show larger fluctuations in amplitude, with the amplitude fluctuations saturating at large window sizes. This saturation of the amplitude variations may contribute to the correlations saturating at larger window sizes.

4. Discussion

We used fine-grained measures of disfluency to explore the determinants of event boundaries in a story-listening task. EST posits that people experience event boundaries at points of transient increase in prediction error (Zacks et al., 2007). In prior work, it has been difficult to measure prediction error at a fine-grained scale to study event cognition. Using GPT-2 (Radford et al., 2019), a predictive deep learning language model, we extracted probability distributions and contextual word embeddings for each word in a narrative. From the probability distributions, we derived the following measures: surprisal, defined as the probability at word-point $t - 1$ of the actual word presented at word-point t ; the entropy of the probability distribution, which gives us a measure of the state of uncertainty about the next word being predicted; and Bayesian surprise, which measures the difference in probability distributions between the current word and the previous word. We also measured the change in contextual embeddings across time points (using cosine distance), as an index of how much the features of the narrative were changing. To test the claim from EST that event boundaries are driven by *transient* changes in prediction error, we normalized the aforementioned measures by a running average, using the formula from Reynolds et al. (2007); we also looked at the measures without this normalization. Our main finding was that—out of all of the disfluency measures we computed—transient Bayesian surprise did the best job of predicting human segmentation behavior. This result is consistent with the EST framework's emphasis on transient prediction error (Zacks et al., 2007) but also refines this idea by showing that a more sophisticated measure of prediction error (Bayesian surprise) outperforms simple surprisal.

4.1. Bayesian surprise: Belief updates, context shifts

What features of Bayesian surprise led to it outperforming the other disfluency measures? Bayesian surprise (Itti & Baldi, 2009) measures the difference between prior and posterior

probability distributions by computing the KLD between these distributions. We can interpret this measure as indexing an update to our priors; the larger the Bayesian surprise, the more our prior expectations were violated. Tracking changes to the entire probability distribution is beneficial in two ways. First, the width of the prior distribution gives us a measure of confidence; the narrower the prior distribution, the more confident we are. Shifts in confidence levels—especially experiencing prediction violations from a confident state—may be indicative of the need to update one's beliefs by loading up a new event model (Dayan & Yu, 2003; Gijssen, Grundei, Lange, Ostwald, & Blankenburg, 2021; Kuperberg, 2021; Kuperberg & Jaeger, 2016; Nassar et al., 2010; Yu & Dayan, 2005). As discussed earlier, Bayesian surprise is more sensitive to the overall shape of the distribution (and thus changes in confidence) than simple surprisal, which only tracks the probability of a single word. Note that Bayesian surprise also performed much better than entropy at predicting human event boundaries; a key difference between these measures is that Bayesian surprise is sensitive to the *change* in confidence across time points, whereas entropy only tracks confidence at a single time point.

A second benefit of tracking the entire probability distribution is that Bayesian surprise can detect changes in the overall content of the narrative. As reviewed, earlier, studies have shown that event boundaries can occur whenever the content of a narrative changes (Huff et al., 2014; Radvansky & Zacks, 2011; Zacks et al., 2007; Zwaan & Radvansky, 1998); furthermore, these content changes can trigger event boundaries even if the changes themselves were not “surprising” in the conventional sense of the word. For example, event boundaries can occur when we anticipate the end of one event and beginning of the other, for example, going to lunch after a class lecture—a “proactive” event boundary (Baldwin & Kosie, 2021; Kuperberg, 2021; Kuperberg & Jaeger, 2016). In this case, the walk towards a restaurant is not unexpected. People still mark event boundaries when they have been explicitly foreshadowed (Pettijohn & Radvansky, 2016) and when they have previously heard a narrative and thus can predict what will happen next (Michelmann et al., 2021). Bayesian surprise can capture this kind of event boundary—the difference in the probability distributions of the words from one context to another (e.g., going from lecture to lunch) will result in a large Bayesian surprise (for additional discussion of how shifts in these kinds of probability distributions can cause event boundaries, see Shin & DuBrow, 2021).

Our embedding cosine distance measure is similar to Bayesian surprise, in that both can track changes in meaning (in fact, they are extracted from adjacent places in the network—the contextual embeddings we use are taken from the last layer before predictions are generated). The embedding cosine distance measure did better overall than surprisal and entropy but discernibly worse than Bayesian surprise. There are two possible reasons for this: First, it is possible that the extra step of “unpacking” the embedding into a predictive distribution is important. Second, it is possible that Bayesian surprise's ability to track overall shifts in confidence (as well as shifts in content) is important; embedding cosine distance does not explicitly track these confidence shifts (although it is possible that it might be correlated with these shifts).

Lastly, it is worth noting that the best-performing version of the Bayesian surprise model was the “transient” version incorporating normalization. This was true even when we used AIC, which penalizes the transient version of the model for having one more free parameter (*drift*) than the non-transient version (Fig. 2). Overall, this supports the claim made by Zacks

et al. (2007) and Reynolds et al. (2007) that prediction errors are more likely to trigger event boundaries when they “stand out” against a background of low prediction error.

4.2. *Limitations of deep language models*

Although deep learning language models have become better at a range of natural language processing tasks compared to their predecessors, they still have limitations in performing tasks at the level of humans. For example, GPT-2 fails at many “system 2” cognitive processing tasks, where we need to carefully reason from a set of premises (Bengio, Lecun, & Hinton, 2021; for limitations of GPT-3, see Binz & Schulz, 2023). These deficits can lead to different probability estimates about the next moment than what humans would estimate.

GPT-2 can only process text input and thus cannot access non-linguistic cues. For example, even without comprehending the semantic content of an auditory narrative, we can use non-linguistic cues such as pauses in speech and changes in intonation to gauge shifts in situations (see Papenmeier, Maurer, & Huff, 2019). GPT-2 also lacks the ability to interpret stutters and voice fillers (“uhms,” “ahs”) that humans can often interpret meaningfully and use to gauge the emotion of the speaker (Hall, Horgan, & Murphy, 2019). If we were to restrict participants to the same level of input that we parse through GPT-2 (i.e., having participants only read text, without auditory input), the model would be on a more even footing with humans and could potentially explain more of the variance in human segmentation; alternatively, we could try to equate the performance of models and humans by drawing on newly developed language models that process actual speech sounds (e.g., Schneider, Baevski, Collobert, & Auli 2019).

4.3. *Testing EST in other modalities*

The work we presented here in the domain of language is closely related to recent work by Fountas et al. (2022) in the visual domain. Instead of using a language model to predict upcoming words, Fountas et al. (2022) used a vision model to predict upcoming information in videos. As in our work, the Fountas et al. (2022) model was used to generate a time series of prediction error using KLD (i.e., what we refer to as Bayesian surprise; for an alternative approach based on feature change, see Roseboom et al. 2019). In their case, rather than predicting human-annotated event boundaries, Fountas et al. (2022) used prediction error to predict human duration judgments for videos. Fountas et al. (2022) first identified putative event boundaries in the video based on the prediction error time series from the model. Next, Fountas et al. (2022) successfully related the number of model-identified event boundaries in the videos to duration judgments made by human participants who watched the videos; this result aligns with previous experimental work linking duration judgments to the number of intervening event boundaries (e.g., Ezzyat & Davachi 2014; Faber & Gennari 2015; Lositsky et al. 2016; Poynter 1983; Zakay, Tsal, Moses, & Shahar 1994). The fact that Bayesian surprise performed well in the Fountas et al. (2022) analyses provides convergent support for the importance of this measure in driving event segmentation.

5. Conclusion

In this study, we leveraged the power of a modern deep learning model of language, GPT-2, to test the central tenets of EST (Zacks et al., 2007). The key benefit of GPT-2 in this context is that it allowed us to estimate fine-grained time series of prediction (and prediction error) for text narratives. Using this GPT-2-derived probability distribution at each moment, we found that the regression models incorporating transient Bayesian surprise did the best job of explaining human segmentation behavior. This result is consistent with EST, which posits that we mark event boundaries when we encounter transient increases in prediction error (Zacks et al., 2007). Importantly, the success of Bayesian surprise in this context (more so than surprisal) suggests a refinement in how we think of the role of prediction error in event segmentation—one where the probability of the observed word matters less than the degree to which the entire predictive distribution has shifted.

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Notes

- 1 One advantage of using GPT-2 over other predictive models is that GPT-2 provides full access to the predictions, probability distributions, and the word embeddings generated by the model. Other models such as GPT-3 or GPT-4 do not presently provide this information. Access to the full GPT-2 model allowed us to compute the disfluency measures that we used in this work.
- 2 In psycholinguistics, surprisal has been defined as being proportional to the negative log probability of the word given the context in which the word appears (e.g., Armeni, Willems, van den Bosch, & Schoffelen, 2019; Levy, 2008; Lowder, Choi, Ferreira, & Henderson, 2018). While our definition of surprisal in terms of the negative log probability of the upcoming word (as in Armeni et al., 2019 and Szewczyk & Federmeier, 2022) does not explicitly condition on context, the model’s computation of surprisal implicitly conditions on context (given that GPT-2 uses context to compute the predictive distribution)—hence it is conceptually aligned with psycholinguistic work on surprisal, even if the mathematical details differ.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1: The figure shows button-press proportion data and the time series for each of the non-transient measures of disfluency (surprisal, entropy, embedding cosine distance, and Bayesian surprise) for each of the stories.

Table S1: Correlation and percentile scores for transient Bayesian surprise for different regression windows and drift values, computed separately for each of the three narratives.

Figure S2: The AIC scores for transient and non-transient disfluency measures, plotted separately for each regression window size.

Table S2: Correlation and percentile scores for transient and non-transient disfluency measures for different regression windows.

Table S3: Correlation and percentile scores for transient Bayesian surprise for different regression windows and drift values, computed separately for each of the three narratives.

Figure S3: The predicted button-press density across different regression window sizes for each of the stories.

The data and code is available at: <https://doi.org/10.5281/zenodo.8404102>